

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 2        | (a) | (i)  | Substitution: $W = \frac{1}{2} \times 1 \times 0.4$ <b>or</b> $W = \frac{1}{2} \times 2.5 \times 0.4^2$ (1)<br>$W = 0.2$ [J] (1)                                                                                                                                                                                                                                                                                                                                                                             | 1               | 1   |     | 2     | 2     |      |
|          |     | (ii) | <b>Either:</b><br>$W = \frac{1}{2}Fx$ (1)<br>$F$ doubles <b>and</b> $x$ doubles <b>or</b> $W$ increased $\times 4$ <b>or</b> increased to 0.8 J (1)<br><b>Or:</b><br>$W = \frac{1}{2}kx^2$ (1)<br>$k$ constant (by implication) and $x$ is squared <b>or</b> $W$ increased $\times 4$ <b>or</b> increased to 0.8 J (1)<br><b>Or:</b><br>Correct use of numerical values from graph e.g.<br>Area under graph stated or implied (1)<br>$W$ increased $\times 4$ <b>or</b> increased to 0.8 J or equivalent (1) | 1               | 1   |     | 2     | 1     |      |

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |            |          |           |          |          |
|----------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|----------|-----------|----------|----------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2        | AO3      | Total     | Maths    | Prac     |
|          | (b) | (i)  | Cons of energy correctly applied (or implied) i.e.<br>$0.20 \text{ (ecf)} = E_k + E_p$ (at top of loop) (1)<br><br>Substitution e.g.<br>$0.20 = \frac{1}{2} \times (30 \times 10^{-3}) \times v^2 + (30 \times 10^{-3}) \times 9.81 \times 0.50$ (1)<br>Award two marks if this seen and first mark not already awarded.<br>$v = 1.8[8 \text{ m s}^{-1}]$ accept $1.9 \text{ m s}^{-1}$ (1)<br>Award 1 mark for $3.7 \text{ m s}^{-1}$ (omitting GPE)<br><br><b>Award 3 marks for the following:</b><br>$\frac{1}{2}mv^2 = 0.2$ leading to $3.65 \text{ [m s}^{-1}]$ (1) can be implied by correct answer<br>Correct application of $v^2 = u^2 + 2as$ i.e. $v^2 = 3.65^2 - 2 \times 9.81 \times 0.5$ (1)<br>Correct answer = $1.87 \text{ [m s}^{-1}]$ (1) |                 | 3          |          | 3         | 3        |          |
|          |     | (ii) | $5\% \times 0.2 = 0.01 \text{ ecf}$ (1)<br>$d$ correct method: $(\pi \times 0.25) + 0.5 = 1.28[5 \text{ m}]$ (1)<br><br>$F = \frac{0.01}{1.285} = 0.0078 \text{ [N]}$ <b>ecf</b> on <b>calculated</b> value of $d$ and <b>calculated</b> value of $E$ (1)<br><br>Accept answers based on $0.9 \text{ m}$ or $0.5 \text{ m}$ i.e. $d = 1.68[5 \text{ m}]$ so $F = 0.006 \text{ [N]}$ so all 3 marks could be awarded                                                                                                                                                                                                                                                                                                                                        | 1               | 1<br><br>1 |          | 3         | 3        |          |
|          |     |      | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>3</b>        | <b>7</b>   | <b>0</b> | <b>10</b> | <b>9</b> | <b>0</b> |